

Supplementary Information

Efficient modes, not load shifting, deliver most of the grid value of flexible AI computing

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Supplementary Information v1.4, 17 September 2026. Base case: idle power 41% of nameplate; 25% is a sensitivity. Metric: total incremental system cost for the same computing work (EUR per representative week, expected over four weeks); reductions are relative to rigid full-speed operation (S0); shifting value is (cost S0e – cost S2)/cost S2. Full-precision, machine-readable versions of every table are provided as Supplementary Data 1 (spreadsheet, one sheet per table) and in the code repository under outputs/research/tables/.

Column key (Tables S7–S12). `cost_X`: incremental system cost under scenario X, i.e. total expected system cost with the AI pool minus that without it (EUR per representative week). `red_A_B_pct` = (1 – cost B / cost A) × 100. `gap_A_B_pct` = (cost A / cost B – 1) × 100. `share_S1` = (cost S0 – cost S1)/(cost S0 – cost S2), the share of total coordination value captured by the tariff-driven firm. `timing_share_X` = (cost S0e – cost X)/(cost S0e – cost S2), the share of shifting value realised by scenario X (negative when shifting raises cost above no-shifting operation). `ai_mwh_X`: AI pool energy under scenario X (MWh per representative week). `new_ocgt_X`, `new_batt_X`: new open-cycle gas-turbine and battery capacity (MW) relative to the reference without AI. `curtail_N0AI`: curtailment share in the reference without AI. `idle`: idle power as a fraction of nameplate. `ai_share`: AI pool as a share of the 2030 provincial peak; `slack_mult`: multiplier on observed queueing wait in the deadline (1 or 3); `export`: interprovincial exchange allowed. In Table S12, `system_cost_X` is the total system cost and savings are in EUR per representative week.

S1. Data registry

id	dataset	url	evidence_type	licence	status	limits
D01	Emerald AI May 2025 experiment	https://github.com/ai-emerald/emerald-ai-demo-may-2025	published_experimental_data	Apache-2.0 repository licence	downloaded_reanalysed	GPU-only modes; timestamp/aggregate conflicts; no independent repeated runs or complete service logs
D02	China 2018 provincial hourly load	https://doi.org/10.5281/zenodo.8322210	reconstructed_historical_load	CC-BY-4.0	checksums_verified_column_mapping_verified	daily-extrema typical-profile reconstruction; not current metered load
D03	China interprovincial transmission	https://doi.org/10.5281/zenodo.8322210	compiled_infrastructure	CC-BY-4.0	quarantined_pending_asset_checks	sheet classifications reversed vs appendix; 3 matrix/list edge conflicts; aliases and endpoints unresolved
D04	PyPSA-China model and assumptions	https://github.com/PyPSA/PyPSA-China	code_and_derived_inputs	MIT code; per-input licences need checking	repository_downloaded_load_provenance_traced	load shares historical origin with D02; not independent validation; default future load scaled shape
D05	Dunlap capacity-adequacy source model	https://github.com/ctdunlap32/data-center-flexibility-resource-adequacy	author_code_manuscript_processed_inputs	MIT code; upstream data licences separate	method_inspected_not_reproduced	exogenous ELCC and partially unobserved assumptions; author manuscript is not Nature publication
D06	Azure LLM inference trace 2024	https://github.com/Azure/AzurePublicDataset/blob/master/AzureLLMInferenceDataset2024.md	production_request_metadata	CC-BY-4.0	schema_license_release_sizes_inspected_not_downloaded	invocation and token lengths only; no deadlines or GPU power; code 692 MB, conversation 1135 MB
A01	Two-block diagnostic scenarios	local analysis	analyst_assumptions	original analysis	implemented_verified	not production workloads or grid results

id	dataset	url	evidence_type	licence	status	limits
A02	Sequential-task diagnostic scenarios	local analysis	analyst_assumptions	original analysis	implemented_verified	uniform arrivals; assumed deadline and idle power; no grid or switching cost
D07	BurstGPT release v2.0 part 1	https://github.com/HPMLL/BurstGPT/releases/tag/v2.0	production_request_metadata	CC-BY-4.0	download_digest_verified_schema_and_hourly_audit_complete	no deadlines, elapsed time, request IDs or power; retain repeated metadata and zero-output records
D08	PyPSA-China V3.0 selected archive members	https://doi.org/10.5281/zenodo.13987282	published_model_inputs	CC-BY-4.0 record; verify upstream embedded sources before redistribution	member_crc_size_verified_and_inputs_audited	not a calibrated base year; source-model totals and official statistics differ; hydro leapday missing; reservoir subsystem separate
D09	2020 national power reliability annual report	https://prpq.nea.gov.cn/uploads/file1/20211009/616107fe94a8e.pdf	official_aggregate_reliability_statistics	public official report; reuse facts with attribution	PDF_obtained_three_tables_extracted_with_sum_checks	not chronological outage telemetry; unplanned/forced/availability metrics differ; unit-year exposure definitions require checking
D10	DynamoLLM HPCA 2025 published experiment	https://jovans2.github.io/files/DynamoLLM_HPCA2025.pdf	published_experimental_methods_and_aggregate_tables	author-hosted paper; numeric facts attributed; no PDF redistribution planned	PDF_obtained_tables_and_methods_inspected	raw power/performance profiling dataset not obtained; cannot treat example SLOs as BurstGPT observed deadlines
A03	Fixed commitment operational cost scenarios	local analysis	analyst_assumptions_with_measured_modes	original analysis	implemented_analytic_dual_and_convexity_checks_passed	assumed tariffs, arrivals, deadlines, idle power; no observed company costs or grid value
D11	GEM Global Integrated Power Tracker July 2025 China subset	https://doi.org/10.5281/zenodo.16810831	compiled_unit_phase_inventory	CC-BY-4.0 confirmed About sheet	download_checksum_verified_unit_ID_and_date_completeness_audited	tracking thresholds, unknown dates, conversions and historical retired coverage require reconciliation; not complete official historical fleet
D12	Alibaba PAI GPU cluster trace 2020 (NSDI 22)	https://github.com/alibaba/clusterdata/tree/master/cluster-trace-gpu-v2020	production_job_task_instance_metadata_and_instance_average_sensors	CC-BY-4.0 (repository LICENSE)	downloaded_extracted_sha256_recorded_audit_complete	no deadlines, no power; plan_gpu is requested not used; wait = scheduling latency only; dates fake
D13	SenseTime Helios GPU cluster traces (SC 21)	https://github.com/S-Lab-System-Group/HeliosData	production_slurm_job_metadata	see LICENSE.txt in release (Apache-2.0 text)	downloaded_sha256_verified_audit_complete	no power or utilisation; queue is observed scheduling wait not a deadline; 39% of GPU-hours in CANCELLED jobs
D14	Microsoft Philly DL training trace (ATC 19)	https://github.com/msr-fiddle/philly-traces	production_job_log_and_per_minute_gpu_utilisation	CC-BY-4.0 (repository LICENSE)	downloaded_via_ifs_sha256_recorded_job_log_and_streamed_utilisation_audit_complete	2017 hardware; per-GPU per-minute utilisation with variable columns (DST duplicate rows truncated to first 8 values); 68% of GPU-hours in Failed/Killed jobs; no deadlines
A04	Region selection matrix	local analysis	screening_table_from_archive_and_official_inputs	original analysis	built	archive capacities unverified; Gansu large hydro absent from archive; Inner Mongolia grid split
A05	Regional S0/S0b/S1/S2 smoke grid	local analysis	uncalibrated_regional_smoke_test	original analysis	pipeline_verified_results_not_empirical	unvalidated hourly load shape; archive capacities; placeholder tariff and external market; no unit commitment or coal minimum load
D15	Jiangsu grid 2020-2022 T&D and retail tariff notice (苏发改价格发〔2020〕1183号) incl. industrial TOU table	https://www.njqxq.gov.cn/qxqrmzf/qxqfzhggj/202011/P020201111396651530459.pdf	official_tariff_document_scanned_pdf	public government document	pdf_saved_pages_read_values_transcribed	scanned images, transcribed by reading; 2021 effective date used for 2020 base year; market users pay market price + T&D

id	dataset	url	evidence_type	licence	status	limits
D16	Gansu 2020-12 notice on retail tariff adjustment and TOU optimisation (policy interpretation page)	http://www.gansu.gov.cn/art/2020/12/3/art_10359_474921.html	official_policy_interpretation_web_page	public government page	mirror_page_fetched_values_transcribed	original gansu.gov.cn page returned HTTP 412 to fetch; mirror gsei.com.cn used; ratios not absolute prices
D17	Guizhou TOU mechanism notice 黔发改价格〔2023〕481号	https://fgw.guizhou.gov.cn/zwgk/zcwj/zcwj/202306/t20230628_80566840.html	official_notice_web_page	public government page	page_fetched_values_transcribed	2023 rules used as nearest official structure; no 2020 Guizhou TOU document obtained; year mismatch flagged
A06	GEM-based unit-level 2020/2030 fleets for Gansu, Jiangsu, Guizhou	local analysis of D11	derived_unit_inventory	CC-BY-4.0 upstream	built_compared_with_archive	GEM size thresholds (lower bound vs official); unknown start years excluded; pre-construction listed separately
A07	Provincial 2030 S0/S1/S2/S3 scenario runs	local analysis	public_data_scenario_with_documented_assumptions	original analysis	pipeline_run_results_labelled_scenario_not_calibrated	hourly load shape unvalidated (2018-derived); external-market proxy; committed-capacity must-run heuristic; RE existing = 2020 archive; no new coal; S3 event rule = >20% above weekly median marginal cost
D18	MLPerf Training v4.0 AC power logs (Supermicro 8xH100, llama2_70b_lora)	https://github.com/mlcommons/training_results_v4.0/tree/main/smc-ac-power	published_benchmark_power_measurements	MLCommons results repository (Apache-2.0)	downloaded_parsed_idle_active_extracted_three_benchmarks	node-level incl. non-GPU components; benchmark workloads; one run per benchmark
D19	Open-Meteo historical weather API (ERA5-based) hourly 2015-2024 at GEM wind/solar sites	https://archive-api.open-meteo.com/v1/archive	reanalysis_weather_free_api	CC-BY 4.0 (Open-Meteo; ERA5 Copernicus)	pulled_cached_converted_to_capacity_factors	simple power-curve and GHI conversions; level scaled to archive 2020 mean; not measured generation

S2. Model verification records

Joint planning–operation LP: 118 checks in total, of which 100 are independent oracle cases (60 screening-curve capacity cases, 40 enumerated task-assignment cases); reservoir-cascade extension: 74 checks, of which 60 are independent dynamic-programming cases; task model: 120 exhaustive assignment oracles and 256 diagnostic cases; commitment-cost model: 72 frontiers, 360 points, dual and finite-difference checks.

S3. MLPerf Training v4.0 node power (8xH100 nodes)

benchmark	nodes	idle_kW	active_kW	max_kW	idle_over_max
llama2_70b_lora	8	2.647	6.25	6.42	0.412
resnet	8	2.24	5.859	5.933	0.378
ssd	8	2.235	6.277	6.372	0.351

S4. GEM-based 2020 and 2030 fleets, three provinces (MW)

province	type	mw_2020	units_2020	mw_2030_operating_plus_construction	units_2030	mw_2030_preconstruction_announced
Gansu	coal	21630	61	33650	75	2000
Gansu	oil/gas	180	1	180	1	0
Gansu	nuclear	0	0	0	0	0
Gansu	hydropower	5514	34	8094	36	0

province	type	mw_2020	units_2020	mw_2030_operating_plus_construction	units_2030	mw_2030_preconstruction_announced
Gansu	wind	12798.4	111	28451.6	215	200
Gansu	solar	8632.1	286	24919.9	407	0
Jiangsu	coal	75804	199	92481	221	2700
Jiangsu	oil/gas	17542	73	24779	108	0
Jiangsu	nuclear	5490	5	9138	8	0
Jiangsu	hydropower	2600	3	3950	5	1200
Jiangsu	wind	13261.2	136	21955.2	179	0
Jiangsu	solar	9303.6	1283	11198	1301	0
Guizhou	coal	31860	75	41890	95	2210
Guizhou	oil/gas	0	0	0	0	0
Guizhou	nuclear	0	0	0	0	0
Guizhou	hydropower	16735	39	18235	40	0
Guizhou	wind	4807.5	95	6626.1	123	100
Guizhou	solar	9095	128	21149.5	236	0

S5. Peak-load plausibility check

province	anchored_2020_peak_MW	official_reference	reference_value_MW	status
Jiangsu	109,828	2020 年夏季与冬季最高调度负荷均突破 1 亿千瓦（国网江苏，2021-01 报道）	>100000	consistent in magnitude; exact value not obtained
Gansu	19679	2021 年全网最大用电负荷 1766 万千瓦（北极星电力网转载）	17660	anchored 2020 peak exceeds 2021 official peak by ~11%
Guizhou	31255	贵州电网统调负荷 2026-01 首次突破 3000 万千瓦（央广网）；2025 年底最高 2955.3 万千瓦	<30000 in 2020	anchored 2020 peak exceeds the 2026 record; 2020 peak must have been well below 31.3 GW

S6. Multi-year renewable profiles: site coverage and agreement with archive 2020 profiles

province_tech	level_scale	corr_with_archive_2020	top_site_capacity_share	n_operating_sites
Gansu / solar	0.955	0.936	0.254	442
Gansu / wind	1.16	0.66	0.303	217
Guizhou / solar	0.711	0.956	0.227	227
Guizhou / wind	2.391	0.954	0.266	125
Jiangsu / solar	0.973	0.967	0.274	1704
Jiangsu / wind	0.901	0.805	0.305	187

S7. 2030 scenario results, islanded (export = False) and fixed-price exchange proxy (export = True), idle 41%

S7a. Incremental system cost by scenario (EUR per representative week)

province	export	ai_share	slack_mult	cost_S0	cost_S0e	cost_S1	cost_S3	cost_S1rt	cost_S2
Gansu	False	0.05	1	4,716,346	4,081,179	4,101,238	4,101,238	4,077,055	4,075,821
Gansu	False	0.05	3	4,716,346	3,757,311	3,781,838	3,781,838	3,753,656	3,753,117
Gansu	False	0.1	1	9,391,522	8,113,527	8,157,286	8,157,286	8,110,767	8,106,916
Gansu	False	0.1	3	9,391,522	7,461,038	7,510,092	7,510,092	7,459,868	7,453,211
Gansu	False	0.2	1	18,494,780	15,929,656	16,017,524	16,017,524	15,924,103	15,917,783
Gansu	False	0.2	3	18,494,780	14,606,454	14,705,016	14,705,016	14,604,649	14,592,950
Gansu	True	0.05	1	5,093,788	4,419,542	4,439,700	4,439,700	4,416,253	4,416,168
Gansu	True	0.05	3	5,093,788	4,070,916	4,095,626	4,095,626	4,068,188	4,068,156
Gansu	True	0.1	1	10,184,505	8,834,452	8,875,171	8,875,171	8,828,409	8,828,313
Gansu	True	0.1	3	10,184,505	8,137,469	8,185,936	8,185,936	8,131,902	8,131,208
Gansu	True	0.2	1	20,352,157	17,651,024	17,731,532	17,731,532	17,638,367	17,638,290
Gansu	True	0.2	3	20,352,157	16,258,282	16,352,857	16,352,857	16,244,905	16,244,482
Guizhou	False	0.05	1	6,406,987	5,585,695	5,557,422	5,578,925	5,517,965	5,503,164
Guizhou	False	0.05	3	6,406,987	5,131,669	5,136,342	5,155,081	5,066,188	5,056,744
Guizhou	False	0.1	1	12,737,351	10,999,516	10,959,127	10,962,125	10,882,236	10,850,800
Guizhou	False	0.1	3	12,737,351	10,074,534	10,089,706	10,118,625	9,972,867	9,937,528
Guizhou	False	0.2	1	25,007,943	21,243,656	21,221,539	21,221,539	21,120,171	20,999,481
Guizhou	False	0.2	3	25,007,943	19,339,271	19,366,970	19,582,761	19,233,822	19,087,095
Guizhou	True	0.05	1	8,264,055	7,163,267	7,198,604	7,198,604	7,159,400	7,159,202
Guizhou	True	0.05	3	8,264,055	6,598,262	6,625,980	6,625,980	6,594,357	6,594,258
Guizhou	True	0.1	1	16,384,621	14,191,353	14,265,554	14,265,554	14,186,077	14,185,069
Guizhou	True	0.1	3	16,384,621	13,065,919	13,125,223	13,125,223	13,061,256	13,060,137
Guizhou	True	0.2	1	32,220,264	27,863,288	28,028,551	28,028,551	27,861,604	27,858,980
Guizhou	True	0.2	3	32,220,264	25,621,877	25,755,850	25,755,850	25,621,578	25,619,232
Jiangsu	False	0.05	1	28,905,044	24,712,571	24,779,466	24,772,013	24,610,641	24,597,050
Jiangsu	False	0.05	3	28,905,044	22,667,442	22,680,700	22,680,700	22,517,787	22,504,657
Jiangsu	False	0.1	1	60,564,006	51,315,483	51,448,812	51,744,240	51,168,598	51,100,438
Jiangsu	False	0.1	3	60,564,006	46,798,303	46,744,502	46,734,867	46,464,921	46,404,857
Jiangsu	False	0.2	1	140,005,403	116,358,518	118,347,943	118,850,619	113,102,184	111,481,799
Jiangsu	False	0.2	3	140,005,403	101,190,396	101,560,918	101,615,317	102,603,733	100,048,295
Jiangsu	True	0.05	1	26,801,003	23,195,195	23,319,371	23,319,371	23,183,959	23,181,802
Jiangsu	True	0.05	3	26,801,003	21,354,049	21,452,284	21,452,284	21,337,164	21,335,366

province	export	ai_share	slack_mult	cost_S0	cost_S0e	cost_S1	cost_S3	cost_S1rt	cost_S2
Jiangsu	True	0.1	1	54,086,719	46,771,340	47,027,907	47,027,907	46,752,940	46,745,368
Jiangsu	True	0.1	3	54,086,719	43,018,531	43,217,892	43,217,892	42,981,138	42,973,250
Jiangsu	True	0.2	1	109,499,873	94,653,841	95,205,981	95,205,981	94,633,015	94,623,243
Jiangsu	True	0.2	3	109,499,873	87,007,025	87,445,255	87,445,255	86,982,402	86,969,565

S7b. Reductions, gaps and realised shares

province	export	ai_share	slack_mult	red_S0_S2_pct	red_S0_S0e_pct	gap_S0e_S2_pct	share_S1	timing_share_S1	timing_share_S1rt	gap_S1rt_S2_pct
Gansu	False	0.05	1	13.581	13.467	0.131	0.96	-3.744	0.77	0.03
Gansu	False	0.05	3	20.423	20.334	0.112	0.97	-5.848	0.872	0.014
Gansu	False	0.1	1	13.678	13.608	0.082	0.961	-6.619	0.417	0.048
Gansu	False	0.1	3	20.639	20.556	0.105	0.971	-6.268	0.15	0.089
Gansu	False	0.2	1	13.934	13.869	0.075	0.961	-7.4	0.468	0.04
Gansu	False	0.2	3	21.097	21.024	0.093	0.971	-7.299	0.134	0.08
Gansu	True	0.05	1	13.303	13.237	0.076	0.965	-5.975	0.975	0.002
Gansu	True	0.05	3	20.135	20.081	0.068	0.973	-8.953	0.989	0.001
Gansu	True	0.1	1	13.316	13.256	0.07	0.965	-6.633	0.984	0.001
Gansu	True	0.1	3	20.161	20.1	0.077	0.973	-7.742	0.889	0.009
Gansu	True	0.2	1	13.335	13.272	0.072	0.966	-6.322	0.994	0
Gansu	True	0.2	3	20.183	20.115	0.085	0.974	-6.853	0.969	0.003
Guizhou	False	0.05	1	14.107	12.819	1.5	0.94	0.343	0.821	0.269
Guizhou	False	0.05	3	21.075	19.905	1.482	0.941	-0.062	0.874	0.187
Guizhou	False	0.1	1	14.811	13.644	1.371	0.943	0.272	0.789	0.29
Guizhou	False	0.1	3	21.981	20.906	1.379	0.946	-0.111	0.742	0.356
Guizhou	False	0.2	1	16.029	15.052	1.163	0.945	0.091	0.506	0.575
Guizhou	False	0.2	3	23.676	22.667	1.321	0.953	-0.11	0.418	0.769
Guizhou	True	0.05	1	13.369	13.32	0.057	0.964	-8.692	0.951	0.003
Guizhou	True	0.05	3	20.206	20.157	0.061	0.981	-6.924	0.975	0.001
Guizhou	True	0.1	1	13.424	13.386	0.044	0.963	-11.809	0.84	0.007
Guizhou	True	0.1	3	20.29	20.255	0.044	0.98	-10.256	0.806	0.009
Guizhou	True	0.2	1	13.536	13.522	0.015	0.961	-38.367	0.391	0.009
Guizhou	True	0.2	3	20.487	20.479	0.01	0.979	-50.657	0.113	0.009
Jiangsu	False	0.05	1	14.904	14.504	0.47	0.958	-0.579	0.882	0.055
Jiangsu	False	0.05	3	22.143	21.58	0.723	0.972	-0.081	0.919	0.058
Jiangsu	False	0.1	1	15.626	15.271	0.421	0.963	-0.62	0.683	0.133
Jiangsu	False	0.1	3	23.379	22.729	0.848	0.976	0.137	0.847	0.129

province	export	ai_share	slack_mult	red_S0_S2_pct	red_S0_S0e_pct	gap_S0e_S2_pct	share_S1	timing_share_S1	timing_share_S1rt	gap_S1rt_S2_pct
Jiangsu	False	0.2	1	20.373	16.89	4.374	0.759	-0.408	0.668	1.453
Jiangsu	False	0.2	3	28.54	27.724	1.142	0.962	-0.324	-1.237	2.554
Jiangsu	True	0.05	1	13.504	13.454	0.058	0.962	-9.272	0.839	0.009
Jiangsu	True	0.05	3	20.393	20.324	0.088	0.979	-5.258	0.904	0.008
Jiangsu	True	0.1	1	13.573	13.525	0.056	0.962	-9.879	0.708	0.016
Jiangsu	True	0.1	3	20.548	20.464	0.105	0.978	-4.403	0.826	0.018
Jiangsu	True	0.2	1	13.586	13.558	0.032	0.961	-18.045	0.681	0.01
Jiangsu	True	0.2	3	20.576	20.541	0.043	0.979	-11.699	0.657	0.015

S7c. AI energy and new capacity

province	export	ai_share	slack_mult	ai_mwh_S0	ai_mwh_S2	new_ocgt_S0	new_ocgt_S2	new_batt_S0
Gansu	False	0.05	1	172,282	149,992	0	0	0
Gansu	False	0.05	3	172,282	138,555	0	0	0
Gansu	False	0.1	1	344,564	299,590	0	0	0
Gansu	False	0.1	3	344,564	276,421	0	0	0
Gansu	False	0.2	1	689,127	598,274	0	0	0
Gansu	False	0.2	3	689,127	551,319	0	0	0
Gansu	True	0.05	1	172,282	149,564	0	0	0
Gansu	True	0.05	3	172,282	137,710	0	0	0
Gansu	True	0.1	1	344,564	299,028	0	0	0
Gansu	True	0.1	3	344,564	275,372	0	0	0
Gansu	True	0.2	1	689,127	597,887	0	0	0
Gansu	True	0.2	3	689,127	550,653	0	0	0
Guizhou	False	0.05	1	289,389	260,341	0	0	0
Guizhou	False	0.05	3	289,389	246,129	0	0	0
Guizhou	False	0.1	1	578,779	517,361	0	0	0
Guizhou	False	0.1	3	578,779	488,041	0	0	0
Guizhou	False	0.2	1	1,157,558	1,022,775	0	0	0
Guizhou	False	0.2	3	1,157,558	958,617	0	0	0
Guizhou	True	0.05	1	289,389	251,072	0	0	0
Guizhou	True	0.05	3	289,389	231,239	0	0	0
Guizhou	True	0.1	1	578,779	502,144	0	0	0
Guizhou	True	0.1	3	578,779	462,478	0	0	0
Guizhou	True	0.2	1	1,157,558	1,004,288	0	0	0
Guizhou	True	0.2	3	1,157,558	924,956	0	0	0

province	export	ai_share	slack_mult	ai_mwh_S0	ai_mwh_S2	new_ocgt_S0	new_ocgt_S2	new_batt_S0
Jiangsu	False	0.05	1	896,015	777,748	0	0	0
Jiangsu	False	0.05	3	896,015	715,968	0	0	0
Jiangsu	False	0.1	1	1,792,029	1,555,066	0	0	165.982
Jiangsu	False	0.1	3	1,792,029	1,431,936	0	0	165.982
Jiangsu	False	0.2	1	3,584,059	3,118,654	9498.094	591.061	969.722
Jiangsu	False	0.2	3	3,584,059	2,870,654	9498.094	0	969.722
Jiangsu	True	0.05	1	896,015	777,375	0	0	0
Jiangsu	True	0.05	3	896,015	715,968	0	0	0
Jiangsu	True	0.1	1	1,792,029	1,554,750	0	0	0
Jiangsu	True	0.1	3	1,792,029	1,431,936	0	0	0
Jiangsu	True	0.2	1	3,584,059	3,109,500	0	0	0
Jiangsu	True	0.2	3	3,584,059	2,863,872	0	0	0

S8. 2030 scenario results, neighbour-aggregate exchange node (export = True rows), idle 41%

S8a. Incremental system cost by scenario (EUR per representative week)

province	ai_share	slack_mult	cost_S0	cost_S0e	cost_S1	cost_S3	cost_S1rt	cost_S2
Gansu	0.05	1	4,790,243	4,160,773	4,179,565	4,179,565	4,155,356	4,155,344
Gansu	0.05	3	4,790,243	3,836,905	3,864,010	3,864,010	3,832,843	3,832,843
Gansu	0.1	1	9,578,732	8,319,781	8,357,807	8,357,807	8,311,059	8,309,365
Gansu	0.1	3	9,578,732	7,672,038	7,726,246	7,726,246	7,663,942	7,663,934
Gansu	0.2	1	19,148,256	16,635,459	16,713,269	16,713,269	16,616,220	16,616,145
Gansu	0.2	3	19,148,256	15,334,546	15,442,977	15,442,977	15,320,840	15,320,114
Guizhou	0.05	1	7,957,847	6,917,865	6,939,766	6,939,766	6,906,688	6,906,486
Guizhou	0.05	3	7,957,847	6,377,242	6,404,402	6,404,402	6,358,127	6,357,912
Guizhou	0.1	1	15,912,304	13,822,922	13,868,325	13,868,325	13,802,506	13,800,894
Guizhou	0.1	3	15,912,304	12,738,025	12,792,399	12,792,399	12,703,082	12,699,252
Guizhou	0.2	1	31,796,779	27,576,531	27,673,179	27,673,179	27,541,847	27,537,910
Guizhou	0.2	3	31,796,779	25,395,318	25,504,306	25,504,306	25,336,671	25,324,864
Jiangsu	0.05	1	25,357,177	21,990,421	22,116,628	22,116,628	21,986,826	21,986,396
Jiangsu	0.05	3	25,357,177	20,250,776	20,352,469	20,352,469	20,246,219	20,245,767
Jiangsu	0.1	1	50,791,070	44,044,923	44,295,969	44,295,969	44,038,246	44,036,425
Jiangsu	0.1	3	50,791,070	40,552,024	40,752,646	40,752,646	40,539,961	40,539,502
Jiangsu	0.2	1	101,894,177	88,322,269	88,822,356	88,822,356	88,307,976	88,302,467
Jiangsu	0.2	3	101,894,177	81,305,696	81,713,472	81,713,472	81,280,684	81,276,226

S8b. Reductions, gaps and realised shares

province	ai_share	slack_mult	red_S0_S2_pct	red_S0_S0e_pct	gap_S0e_S2_pct	share_S1	timing_share_S1	timing_share_S1rt	gap_S1rt_S2_pct
Gansu	0.05	1	13.254	13.141	0.131	0.962	-3.461	0.998	0
Gansu	0.05	3	19.986	19.902	0.106	0.967	-6.672	1	0
Gansu	0.1	1	13.252	13.143	0.125	0.962	-3.651	0.837	0.02
Gansu	0.1	3	19.99	19.905	0.106	0.967	-6.69	0.999	0
Gansu	0.2	1	13.224	13.123	0.116	0.962	-4.029	0.996	0
Gansu	0.2	3	19.992	19.917	0.094	0.968	-7.513	0.95	0.005
Guizhou	0.05	1	13.212	13.069	0.165	0.968	-1.925	0.982	0.003
Guizhou	0.05	3	20.105	19.862	0.304	0.971	-1.405	0.989	0.003
Guizhou	0.1	1	13.269	13.131	0.16	0.968	-2.061	0.927	0.012
Guizhou	0.1	3	20.192	19.949	0.305	0.971	-1.402	0.901	0.03
Guizhou	0.2	1	13.394	13.273	0.14	0.968	-2.502	0.898	0.014
Guizhou	0.2	3	20.354	20.132	0.278	0.972	-1.547	0.832	0.047
Jiangsu	0.05	1	13.293	13.277	0.018	0.961	-31.358	0.893	0.002
Jiangsu	0.05	3	20.158	20.138	0.025	0.979	-20.303	0.91	0.002
Jiangsu	0.1	1	13.299	13.282	0.019	0.962	-29.539	0.786	0.004
Jiangsu	0.1	3	20.184	20.159	0.031	0.979	-16.022	0.963	0.001
Jiangsu	0.2	1	13.339	13.32	0.022	0.962	-25.254	0.722	0.006
Jiangsu	0.2	3	20.235	20.206	0.036	0.979	-13.837	0.849	0.005

S9. Multi-weather-year summary (islanded, export = False, AI 10%; min/median/max across ten weather years, 2015–2024)

Rows are metric_statistic; columns are province / idle-power fraction.

metric	Gansu / idle 0.25	Gansu / idle 0.41	Guizhou / idle 0.25	Guizhou / idle 0.41	Jiangsu / idle 0.25	Jiangsu / idle 0.41
red_S0_S2_pct_min	15.185	13.484	15.672	14.125	15.743	14.389
red_S0_S2_pct_median	15.442	13.922	16.922	14.833	16.269	14.92
red_S0_S2_pct_max	16.063	14.305	17.48	15.32	16.86	15.536
red_S0_S0e_pct_min	14.598	13.159	14.051	13.289	15.136	13.954
red_S0_S0e_pct_median	15.035	13.621	14.722	13.776	15.727	14.513
red_S0_S0e_pct_max	15.227	13.809	15.108	13.964	16.189	15.09
gap_S0e_S2_pct_min	0.143	0.102	1.922	0.968	0.513	0.41
gap_S0e_S2_pct_median	0.673	0.285	2.703	1.246	0.791	0.511
gap_S0e_S2_pct_max	1.073	0.685	3.296	1.795	0.948	0.664
timing_share_S1_min	-6.288	-5.091	0.191	0.039	-1.885	-0.988
timing_share_S1_median	-0.469	-1.437	0.312	0.234	-0.989	-0.691

metric	Gansu / idle 0.25	Gansu / idle 0.41	Guizhou / idle 0.25	Guizhou / idle 0.41	Jiangsu / idle 0.25	Jiangsu / idle 0.41
timing_share_S1_max	-0.051	0.006	0.409	0.334	-0.355	-0.404
gap_S1_S2_pct_min	0.931	0.614	1.377	0.763	1.095	0.74
gap_S1_S2_pct_median	1.114	0.687	1.749	0.957	1.481	0.843
gap_S1_S2_pct_max	1.436	0.805	2.356	1.465	1.934	1.219
gap_S1rt_S2_pct_min	0.058	0.012	0.364	0.187	0.168	0.022
gap_S1rt_S2_pct_median	0.125	0.069	0.569	0.284	0.258	0.071
gap_S1rt_S2_pct_max	0.389	0.215	0.784	0.354	0.31	0.173
curtail_NOAI_min	0.036	0.036	0.122	0.122	0	0
curtail_NOAI_median	0.055	0.055	0.249	0.249	0	0
curtail_NOAI_max	0.105	0.105	0.322	0.323	0	0
new_ocgt_S0_min	0	0	0	0	0	0
new_ocgt_S0_median	0	0	0	0	0	0
new_ocgt_S0_max	0	0	0	0	0	0
new_ocgt_S2_min	0	0	0	0	0	0
new_ocgt_S2_median	0	0	0	0	0	0
new_ocgt_S2_max	0	0	0	0	0	0

S10. Cost-parameter Monte Carlo summary (islanded, export = False, AI 10%; 20 feasible draws per province, infeasible or timed-out draws excluded)

Rows are metric and statistic (count, mean, standard deviation, minimum, 10th/50th/90th percentile, maximum); columns are provinces.

metric	statistic	Gansu	Guizhou	Jiangsu
red_S0_S2_pct	count	20	20	20
red_S0_S2_pct	mean	19.592	18.259	18.329
red_S0_S2_pct	std	5.983	6.947	6.705
red_S0_S2_pct	min	8.785	10.071	9.001
red_S0_S2_pct	10%	9.543	10.835	10.286
red_S0_S2_pct	50%	21.229	17.416	19.31
red_S0_S2_pct	90%	24.802	27.05	26.704
red_S0_S2_pct	max	29.091	28.22	27.158
red_S0_S0e_pct	count	20	20	20
red_S0_S0e_pct	mean	19.196	16.895	17.767
red_S0_S0e_pct	std	5.806	6.673	6.57
red_S0_S0e_pct	min	8.742	9.345	8.998
red_S0_S0e_pct	10%	9.412	9.658	10.009

metric	statistic	Gansu	Guizhou	Jiangsu
red_S0_S0e_pct	50%	20.793	15.666	18.044
red_S0_S0e_pct	90%	24.293	25.768	25.923
red_S0_S0e_pct	max	28.105	25.991	26.305
gap_S0e_S2_pct	count	20	20	20
gap_S0e_S2_pct	mean	0.512	1.708	0.706
gap_S0e_S2_pct	std	0.587	0.704	0.479
gap_S0e_S2_pct	min	0.047	0.807	0.003
gap_S0e_S2_pct	10%	0.079	0.936	0.235
gap_S0e_S2_pct	50%	0.33	1.625	0.678
gap_S0e_S2_pct	90%	1	2.536	1.146
gap_S0e_S2_pct	max	2.657	3.419	2.023
timing_share_S1	count	20	20	20
timing_share_S1	mean	-1.544	0.253	-3.289
timing_share_S1	std	2.485	0.186	8.396
timing_share_S1	min	-7.774	-0.169	-36.07
timing_share_S1	10%	-4.565	-0.017	-7.205
timing_share_S1	50%	-0.142	0.28	-0.165
timing_share_S1	90%	0.426	0.412	0.236
timing_share_S1	max	0.569	0.534	0.337
gap_S1_S2_pct	count	20	20	20
gap_S1_S2_pct	mean	0.766	1.325	1.131
gap_S1_S2_pct	std	0.697	0.799	0.811
gap_S1_S2_pct	min	0.032	0.435	0.093
gap_S1_S2_pct	10%	0.234	0.665	0.3
gap_S1_S2_pct	50%	0.481	1.195	0.933
gap_S1_S2_pct	90%	1.503	1.9	2.37
gap_S1_S2_pct	max	2.628	3.998	2.964
gap_S1rt_S2_pct	count	20	20	20
gap_S1rt_S2_pct	mean	0.118	0.464	0.189
gap_S1rt_S2_pct	std	0.112	0.275	0.16
gap_S1rt_S2_pct	min	0.022	0.091	0
gap_S1rt_S2_pct	10%	0.034	0.177	0.03
gap_S1rt_S2_pct	50%	0.087	0.412	0.146
gap_S1rt_S2_pct	90%	0.202	0.783	0.349
gap_S1rt_S2_pct	max	0.518	1.08	0.586

metric	statistic	Gansu	Guizhou	Jiangsu
new_ocgt_S0	count	20	20	20
new_ocgt_S0	mean	0	0	0
new_ocgt_S0	std	0	0	0
new_ocgt_S0	min	0	0	0
new_ocgt_S0	10%	0	0	0
new_ocgt_S0	50%	0	0	0
new_ocgt_S0	90%	0	0	0
new_ocgt_S0	max	0	0	0
new_ocgt_S2	count	20	20	20
new_ocgt_S2	mean	0	0	0
new_ocgt_S2	std	0	0	0
new_ocgt_S2	min	0	0	0
new_ocgt_S2	10%	0	0	0
new_ocgt_S2	50%	0	0	0
new_ocgt_S2	90%	0	0	0
new_ocgt_S2	max	0	0	0

S11. Peak-adjusted sensitivity (incremental cost, EUR per representative week)

province	ext_mode	export	peak_adjusted	ai_share	case	inc_cost	new_ocgt	new_batt	curtail_NOAI	peak_2030
Gansu	price	False	False	0.1	S0	9,391,522	0	0	0.025	24971.916
Gansu	price	False	False	0.1	S0e	8,113,527	0	0	0.025	24971.916
Gansu	price	False	False	0.1	S1	8,157,286	0	0	0.025	24971.916
Gansu	price	False	False	0.1	S1rt	8,110,767	0	0	0.025	24971.916
Gansu	price	False	False	0.1	S2	8,106,916	0	0	0.025	24971.916
Gansu	price	False	False	0.1	S3	8,157,286	0	0	0.025	24971.916
Gansu	price	False	False	0.2	S0	18,494,780	0	0	0.068	24971.916
Gansu	price	False	False	0.2	S0e	15,929,656	0	0	0.068	24971.916
Gansu	price	False	False	0.2	S1	16,017,524	0	0	0.068	24971.916
Gansu	price	False	False	0.2	S1rt	15,924,103	0	0	0.068	24971.916
Gansu	price	False	False	0.2	S2	15,917,783	0	0	0.068	24971.916
Gansu	price	False	False	0.2	S3	16,017,524	0	0	0.068	24971.916
Gansu	price	False	True	0.1	S0	8,488,611	0	0	0.017	22409.549
Gansu	price	False	True	0.1	S0e	7,340,650	0	0	0.017	22409.549
Gansu	price	False	True	0.1	S1	7,381,484	0	0	0.017	22409.549
Gansu	price	False	True	0.1	S1rt	7,338,515	0	0	0.017	22409.549

province	ext_mode	export	peak_adjusted	ai_share	case	inc_cost	new_ocgt	new_batt	curtail_NOAI	peak_2030
Gansu	price	False	True	0.1	S2	7,336,706	0	0	0.017	22409.549
Gansu	price	False	True	0.1	S3	7,381,484	0	0	0.017	22409.549
Gansu	price	False	True	0.2	S0	16,718,273	0	0	0.052	22409.549
Gansu	price	False	True	0.2	S0e	14,414,338	0	0	0.052	22409.549
Gansu	price	False	True	0.2	S1	14,497,019	0	0	0.052	22409.549
Gansu	price	False	True	0.2	S1rt	14,414,250	0	0	0.052	22409.549
Gansu	price	False	True	0.2	S2	14,408,208	0	0	0.052	22409.549
Gansu	price	False	True	0.2	S3	14,497,019	0	0	0.052	22409.549
Gansu	neighbours	True	False	0.1	S0	9,578,732	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.1	S0e	8,319,781	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.1	S1	8,357,807	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.1	S1rt	8,311,059	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.1	S2	8,309,365	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.1	S3	8,357,807	0	0	0.004	24971.916
Gansu	neighbours	True	False	0.2	S0	19,148,256	0	0	0.006	24971.916
Gansu	neighbours	True	False	0.2	S0e	16,635,459	0	0	0.006	24971.916
Gansu	neighbours	True	False	0.2	S1	16,713,269	0	0	0.006	24971.916
Gansu	neighbours	True	False	0.2	S1rt	16,616,220	0	0	0.006	24971.916
Gansu	neighbours	True	False	0.2	S2	16,616,145	0	0	0.006	24971.916
Gansu	neighbours	True	False	0.2	S3	16,713,269	0	0	0.006	24971.916
Gansu	neighbours	True	True	0.1	S0	8,597,494	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.1	S0e	7,467,710	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.1	S1	7,501,527	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.1	S1rt	7,458,148	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.1	S2	7,458,068	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.1	S3	7,501,527	0	0	0.003	22409.549
Gansu	neighbours	True	True	0.2	S0	17,186,950	0	0	0.005	22409.549
Gansu	neighbours	True	True	0.2	S0e	14,931,508	0	0	0.005	22409.549
Gansu	neighbours	True	True	0.2	S1	15,001,276	0	0	0.005	22409.549
Gansu	neighbours	True	True	0.2	S1rt	14,914,691	0	0	0.005	22409.549
Gansu	neighbours	True	True	0.2	S2	14,914,307	0	0	0.005	22409.549
Gansu	neighbours	True	True	0.2	S3	15,001,276	0	0	0.005	22409.549
Guizhou	price	False	False	0.1	S0	12,737,351	0	0	0.306	41946.452
Guizhou	price	False	False	0.1	S0e	10,999,516	0	0	0.306	41946.452
Guizhou	price	False	False	0.1	S1	10,959,127	0	0	0.306	41946.452

province	ext_mode	export	peak_adjusted	ai_share	case	inc_cost	new_ocgt	new_batt	curtail_NOAI	peak_2030
Guizhou	price	False	False	0.1	S1rt	10,882,236	0	0	0.306	41946.452
Guizhou	price	False	False	0.1	S2	10,850,800	0	0	0.306	41946.452
Guizhou	price	False	False	0.1	S3	10,962,125	0	0	0.306	41946.452
Guizhou	price	False	False	0.2	S0	25,007,943	0	0	0.435	41946.452
Guizhou	price	False	False	0.2	S0e	21,243,656	0	0	0.435	41946.452
Guizhou	price	False	False	0.2	S1	21,221,539	0	0	0.435	41946.452
Guizhou	price	False	False	0.2	S1rt	21,120,171	0	0	0.435	41946.452
Guizhou	price	False	False	0.2	S2	20,999,481	0	0	0.435	41946.452
Guizhou	price	False	False	0.2	S3	21,221,539	0	0	0.435	41946.452
Guizhou	price	False	True	0.1	S0	12,739,894	0	0	0.255	38919.945
Guizhou	price	False	True	0.1	S0e	11,054,777	0	0	0.255	38919.945
Guizhou	price	False	True	0.1	S1	11,029,002	0	0	0.255	38919.945
Guizhou	price	False	True	0.1	S1rt	10,963,723	0	0	0.255	38919.945
Guizhou	price	False	True	0.1	S2	10,931,892	0	0	0.255	38919.945
Guizhou	price	False	True	0.1	S3	11,029,002	0	0	0.255	38919.945
Guizhou	price	False	True	0.2	S0	24,696,485	0	0	0.401	38919.945
Guizhou	price	False	True	0.2	S0e	21,004,844	0	0	0.401	38919.945
Guizhou	price	False	True	0.2	S1	21,051,233	0	0	0.401	38919.945
Guizhou	price	False	True	0.2	S1rt	20,941,978	0	0	0.401	38919.945
Guizhou	price	False	True	0.2	S2	20,834,107	0	0	0.401	38919.945
Guizhou	price	False	True	0.2	S3	21,051,233	0	0	0.401	38919.945
Guizhou	neighbours	True	False	0.1	S0	15,912,304	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.1	S0e	13,822,922	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.1	S1	13,868,325	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.1	S1rt	13,802,506	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.1	S2	13,800,894	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.1	S3	13,868,325	0	0	0.001	41946.452
Guizhou	neighbours	True	False	0.2	S0	31,796,779	0	0	0.003	41946.452
Guizhou	neighbours	True	False	0.2	S0e	27,576,531	0	0	0.003	41946.452
Guizhou	neighbours	True	False	0.2	S1	27,673,179	0	0	0.003	41946.452
Guizhou	neighbours	True	False	0.2	S1rt	27,541,847	0	0	0.003	41946.452
Guizhou	neighbours	True	False	0.2	S2	27,537,910	0	0	0.003	41946.452
Guizhou	neighbours	True	False	0.2	S3	27,673,179	0	0	0.003	41946.452
Guizhou	neighbours	True	True	0.1	S0	14,827,909	0	0	0	38919.945
Guizhou	neighbours	True	True	0.1	S0e	12,874,986	0	0	0	38919.945

province	ext_mode	export	peak_adjusted	ai_share	case	inc_cost	new_ocgt	new_batt	curtail_NOAI	peak_2030
Guizhou	neighbours	True	True	0.1	S1	12,919,408	0	0	0	38919.945
Guizhou	neighbours	True	True	0.1	S1rt	12,860,265	0	0	0	38919.945
Guizhou	neighbours	True	True	0.1	S2	12,857,139	0	0	0	38919.945
Guizhou	neighbours	True	True	0.1	S3	12,919,408	0	0	0	38919.945
Guizhou	neighbours	True	True	0.2	S0	29,618,173	0	0	0.001	38919.945
Guizhou	neighbours	True	True	0.2	S0e	25,683,631	0	0	0.001	38919.945
Guizhou	neighbours	True	True	0.2	S1	25,781,296	0	0	0.001	38919.945
Guizhou	neighbours	True	True	0.2	S1rt	25,653,364	0	0	0.001	38919.945
Guizhou	neighbours	True	True	0.2	S2	25,652,339	0	0	0.001	38919.945
Guizhou	neighbours	True	True	0.2	S3	25,781,296	0	0	0.001	38919.945

S12. Mechanism comparison

S12a. Settlement designs under identical tasks and reliability (rows are metrics; columns are provinces)

metric	Gansu	Jiangsu	Guizhou
system_cost_S1	159,260,672	550,965,702	114,426,853
system_cost_S3	159,260,672	551,261,129	114,429,851
system_cost_S1rt	159,214,152	550,685,488	114,349,963
system_cost_S2	159,210,301	550,617,327	114,318,526
s3_saving_vs_S1	0	-295,427	-2998.019
s3_compensation_floor_own_baseline	0	459,446	20236.509
s3_mean_commitment_mw	0	295.466	41.165
s1rt_saving_vs_S1	46519.224	280,214	76890.129
gap_S1rt_S2_pct	0.048	0.133	0.29
system_cost_S0	160,494,908	560,080,895	116,205,077
system_cost_S0e	159,216,912	550,832,373	114,467,242
system_cost_NOAI	151,103,385	499,516,889	103,467,726
n_events	{'winter': 0, 'spring': 0, 'summer': 0, 'autumn': 0}	{'winter': 8, 'spring': 0, 'summer': 0, 'autumn': 0}	{'winter': 0, 'spring': 0, 'summer': 8, 'autumn': 0}
ai_mwh_S1	300,635	1,563,793	505,168
ai_mwh_S1rt	299,660	1,555,012	518,678

S12b. Baseline-manipulation exposure of the event contract (rows are metrics; columns are province / week)

metric	Guizhou / week summer	Jiangsu / week winter
n_events	8	8
commit_own_mw	164.662	1181.866

metric	Guizhou / week summer	Jiangsu / week winter
comp_own	80946.034	1,837,782
commit_inflated_mw	51.262	0
comp_inflated	16112.12	0
inflated_minus_own_event_power_mw	689.945	3708.718